

Accessible Aerospace Packaging Design

Design protective packaging that can be opened by a user with limited hand strength.

Customer Skyline Aircraft Support	Department Packaging Engineering	Difficulty Intermediate	Time 2-3 class periods
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Production Goal: Protect a sensor while improving accessibility.

Background

A maintenance kit for aircraft technicians contains a fragile sensor. Current packaging protects the part but is difficult to open while wearing gloves or for users with limited grip strength. Your team must redesign the package without reducing protection.

Learning Objectives

- Identify user and product requirements
- Prototype accessible packaging features
- Balance protection, material use, and usability
- Conduct a repeatable user test

Equipment

- Sample component or dimensional model
- Cardboard/paperboard/foam and approved materials
- Cutting and measuring tools
- Scale
- User test rubric
- Engineering notebook

Student Instructions

- 1 Measure the component and document fragile surfaces.
- 2 Interview or use the provided user profile to identify opening challenges.
- 3 Define measurable requirements for protection, opening force, material, and size.
- 4 Sketch three concepts with different opening methods.
- 5 Use a decision matrix to select one concept.
- 6 Build a full-size prototype.
- 7 Conduct a protection test and an opening test with at least three users or simulated conditions.
- 8 Record time, errors, and user feedback.
- 9 Revise one feature and retest.

Engineering Requirements

- Protect the component during the approved drop/handling test
- Open in under 20 seconds without tools
- Use clear opening cues
- Use no more than the approved material allowance
- Allow the package to close securely for transport

Constraints & Safety

- Use only approved safe materials and cutting tools
- Do not expose users to sharp staples or edges
- Do not reveal the component during the protection test
- Collect anonymous usability feedback

Deliverables

- User-needs summary
- Three concept sketches
- Decision matrix
- Prototype
- Protection and usability data
- Revision evidence
- Reflection

Reflection Questions

- 1 Which accessibility feature helped most?
- 2 What tradeoff occurred between security and ease of opening?
- 3 How did user data change the design?
- 4 How could packaging be made more sustainable?

Engineering Support

Engineering Tip

Use large tabs, clear grip zones, and obvious opening direction rather than relying only on text instructions.

Common Mistake

Designing for the average user can exclude technicians wearing gloves or users with limited dexterity.

Stretch Goal

Redesign the package for one-handed opening or reusable return shipping.

Career Connection

Packaging engineers combine material science, logistics, accessibility, branding, and product protection.

Before You Submit

- Notebook evidence included
- Requirements checked
- Testing documented
- Reflection completed